

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10							*****VALVES*****			814	NIPPLE TO BE CUT FROM SS SCH 160 PE PIPE.
20										816	COMPACT FLANGES CAN BE USED FOR TIE-IN PURPOSE ONLY TO EXISTING PIPING AND EQUIPMENT.
20							***** MODULAR VALVES *****				
30										817	USE ONLY FOR TIE IN
30		VA5	MODULAR DBB	K15AL10	0.5	1.5	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 625 INC 625 TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	B1	ALL BOLTS ARE METRIC DIA.
40			MODULAR DBB	K15AL14	2	2	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 625 INC 625 TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10
50		VAW, 817	MODULAR DBB	K15CL14	3	3	MODULAR VALVE DB&B BALL TYPE 1500# CF INC 625 INC 625 TRIM METAL LO - FULL SIZE TYPE 2BV	0	0		
60			MODULAR DBB	M15AL10	0.75	1.5	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 625 INC 625 TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	BA1	TO BE USED WITH RAISED FACE FLANGES ONLY.
70			MODULAR DBB	M15AL14	2	2	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 625 INC 625 TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	BA4	TO BE USED WITH COMPACT FLANGES REQUIRING STANDARD NUTS ONLY.
80	U04	VAW	MODULAR DBB	M15EL10	0.75	1	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 625 INC 625 TRIM METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR	2	2	FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.
90	U04	VAW, 817	MODULAR DBB	M15ZL10	1	1	MODULAR VALVE DB&B BALL TYPE 1500# CF X RF INC 625 INC 625 TRIM METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR	3	3	UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.
100	U04	VAW, 817	MODULAR DBB	K15ML14	3	3	MODULAR VALVE DB&B BALL TYPE 1500# CF X RF INC 625 INC 625 TRIM METAL LO - FULL SIZE TYPE 2BV	0	0		
110	U04	VAW	MODULAR DBB	M15EL14	2	2	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 625 INC 625 TRIM METAL LO REDUCING - TYPE R2BVR	3	3	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
120		VB2					***** BALL VALVES *****			VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.
130											
130			BALL	A15AL13	2	2	BALL 1500# RF INCONEL 625 BODY AND TRIM TRUNNION SOFT RB LO	0	0		
140	U04	817	BALL	B15CL19	3	3	BALL 1500# CF INCONEL 625 BODY AND TRIM TRUNNION SOFT FB GO EXT BONNET	0	0	VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.
150			BALL	A15AL14	2	2	BALL 1500# RF INCONEL 625 BODY AND TRIM TRUNNION METAL RB LO	0	0		
160	U04	817	BALL	A15CL13	3	3	BALL 1500# CF INCONEL 625 TRIM AND BODY TRUNNION SOFT RB LO	0	0	VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.
170	U04	817	BALL	A15CL18	3	3	BALL 1500# CF INCONEL 625 TRIM AND BODY TRUNNION SOFT RB LO EXT BONNET	0	0		
180	U04		BALL	A15AL18	2	2	BALL 1500# RF INCONEL 625 BODY AND TRIM TRUNNION SOFT RB LO EXT BONNET	0	0		
190	U04	817	BALL	A15CL14	3	3	BALL 1500# CF INCONEL 625 TRIM AND BODY TRUNNION METAL RB LO	0	0	VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.
200			BALL	B15AL09	0.75	1.5	BALL 1500# RF INCONEL 625 BODY AND TRIM FLOATING SOFT FB LO	0	0		
210			BALL	B15AL10	0.75	1.5	BALL 1500# RF INCONEL 625 BODY AND TRIM FLOATING METAL FB LO	0	0		
220		VA5	BALL	B15AL13	2	3	BALL 1500# RF INCONEL 625 BODY AND TRIM TRUNNION SOFT FB LO	0	0		
230	U04	VA5	BALL	B15AL18	2	2	BALL 1500# RF INCONEL 625 BODY AND TRIM TRUNNION SOFT FB LO EXT BONNET	0	0		
240		VA5	BALL	B15AL14	2	3	BALL 1500# RF INCONEL 625 BODY AND TRIM TRUNNION METAL FB LO	0	0		
250	U04	817	BALL	B15CL15	4	4	BALL 1500# CF INCONEL 625 BODY AND TRIM TRUNNION SOFT FB GO	0	0		
260	U04	817	BALL	B15CL16	3	4	BALL 1500# CF INCONEL 625 BODY AND TRIM TRUNNION METAL FB GO	0	0		
270							***** GLOBE VALVES *****				
270											
280											
280			GLOBE	L15AL43	0.75	2	GLOBE 1500# RF INC 625 B564 INC 625 TRIM OS&Y HW	0	0		
290	U04	817	GLOBE	L15CL43	3	3	GLOBE 1500# CF INC 625 B564 INC 625 TRIM OS&Y HW	0	0		
300							***** CHECK VALVES *****				
300											
310											
310		VC1	CHECK	C15AL51	0.75	1.5	CHECK PISTON 1500# RF INC 625 TRIM 625	0	0		
320			CHECK	C15AL46	2	2	CHECK DUAL WAFER LUG 1500# RF INC 625 TRIM 625 THROU BOLTED	0	0		
330	U04	817	CHECK	C15CL53	3	3	CHECK DUAL DBL FLG 1500# CF INC 625 TRIM 625	0	0		
340							***** PIPE *****				
340											
350											
350			PIPE	5362274	0.75	1	PIPE SCH 80S SMLS ALLOY 625 B444 UNS N06625 PE	0	0		
360			PIPE	5362270	1.5	1.5	PIPE SCH 40S SMLS ALLOY 625 B444 UNS N06625 PE	0	0		
370	U04		PIPE	5362280	2	4	PIPE SCH 40S SMLS ALLOY 625 B444 UNS N06625 BE	0	0		
380		814	PIPE	5362289	0.75	0.75	PIPE SCH 160 SMLS ALLOY 625 B444 UNS N06625 BE	0	0		
390	U04		PIPE	5362284	6	8	PIPE SCH 80S SMLS ALLOY 625 B444 UNS N06625 BE	0	0		
400							***** FITTINGS *****				
400											
410											
410		UJ2	ELBOW	5426552	0.75	8	ELL BW 90 DEG LR SCH 80S ALLOY 625 B366-WPS UNS N06625	0	0		
420	U04		ELBOW	5426511	1.5	4	ELL BW 90 DEG LR SCH 40S ALLOY 625 B366-WPS UNS N06625	0	0		
430			ELBOW	5426557	0.75	1	ELL BW 45 DEG LR SCH 80S ALLOY 625 B366-WPS UNS N06625	0	0		
440			ELBOW	5426516	1.5	3	ELL BW 45 DEG LR SCH 40S ALLOY 625 B366-WPS UNS N06625	0	0		
450											
450			TEE	5426560	0.75	1	TEE BW SCH 80S ALLOY 625 B366-WPS UNS N06625	0	0		
460			TEE	5426519	1.5	3	TEE BW SCH 40S ALLOY 625 B366-WPS UNS N06625	0	0		
470	U04		TEE RED	5426559	0.75	0.75	TEE RED SCH 80S ALLOY 625 B366-WPS UNS N06625	1	1		
480			FLANGE	5426558	0.75	1	TEE RED SCH 40S/80S ALLOY 625 B366-WPS UNS N06625	1.5	2		
490			FLANGE	5426518	1.5	2	TEE RED SCH 40S ALLOY 625 B366-WPS UNS N06625	2	3		
500											
500			CAP	5426561	0.75	1	CAP SCH 80S ALLOY 625 B366 UNS N06625	0	0		
510			CAP	5426520	1.5	3	CAP SCH 40S ALLOY 625 B366 UNS N06625	0	0		
520											
520			RED CONC	5426562	0.75	0.75	REDUCER CONC SCH 80S ALLOY 625 B366-WPS N06625	1	1		

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD	CORR. ALLOW. 1 MM	SOUR SERVICE NACE STD. MR0175 / ISO 15156 DESIGN: 184 BARG AT 110°C CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: INCONEL 625 FLANGE RATING: 1500 RF/CF TEMPERATURE RANGE: -100 / 110 °C	12/16/2024	BY OZAC	U04	Page 1 of 2	1500A60

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
530			RED CONC	5426530	0.75	1	REDUCER CONC SCH 40S/80S ALLOY 625 B366-WPS N06625	1.5	2	
540			RED CONC	5426521	1.5	3	REDUCER CONC SCH 40S ALLOY 625 B366-WPS N06625	2	3	
550	U04		RED CONC	5425142	4	4	REDUCER CONC SCH 80S/40S ALLOY 825 B366-WPS N08825	8	8	
560			RED ECC	5426563	0.75	0.75	REDUCER ECC SCH 80S ALLOY 625 B366-WPS N06625	1	1	
570			RED ECC	5226531	0.75	1	REDUCER ECC SCH 40S/80S ALLOY 625 B366-WPS N06625	1.5	2	
580	U04		RED ECC	5426522	1.5	3	REDUCER ECC SCH 40S ALLOY 625 B366-WPS N06625	2	4	
590	U04		RED ECC	5426532	4	4	REDUCER ECC SCH 80S/40S ALLOY 625 B366-WPS N06625	8	8	
600										
600							***** OLETS*****			
610										
610			OLETS	5513582	0.75	1	NIPOFLANGE 1500# RF SCH 40S/80S ALLOY 625 B564 UNS N06625	2	3	
620			OLETS	5513580	1.5	1.5	NIPOFLANGE 1500# RF SCH 40S ALLOY 625 B564 UNS N06625	2	3	
630		UJ2	OLETS	5516681	1.5	1.5	NIPOFLANGE 1500# RF SCH 40S/160 ALLOY 625 B564 UNS N06625	3	3	
640										
640			OLETS	5513556	0.75	1	WELDOLET SCH 40S/80S ALLOY 625 B564 UNS N06625	2	3	
650										
650		UJ2	ELBOFLG	5516181	1.5	1.5	ELBOFLANGE 1500# RF SCH 40S/160 ALLOY 625 B564 UNS N06625	3	3	
660										
660		816					***** FLANGES*****			
670										
670		FA8	FLANGE	5614410	0.5	0.5	FLG SCRDR (FNPT) REDUCING 1500# RF ALLOY 625 B564 UNS N06625	0.75	1.5	
680										
680			FLG ADAPTER	5851215	0.75	2	FLANGE ADAPTER 1500# RF ALLOY 625 B564 UNS N06625 X 12 MM TUBE	0	0	
690										
690			FLANGE	5571900	0.75	1	FLG WN 1500# RF ALLOY 625 SCH 80S B564 UNS N06625	0	0	
700		816	FLANGE	5571901	1.5	3	FLG WN 1500# RF ALLOY 625 SCH 40S B564 UNS N06625	0	0	
710		FA8	BLIND	5617186	0.5	3	FLG BLIND 1500# RF ALLOY 625 B564 UNS N06625	0	0	
720	U04	817	FLANGE	5619849	3	3	FLG COMPACT 1500# RJ ALLOY 625 SCH 40S B564 UNS N06625	0	0	
730	U04	817	FLANGE	5563369	8	8	FLG COMPACT 1500# RJ ALLOY 625 SCH 80S B564 UNS N06625 LINE CLASS 1500A60	0	0	
740	U04	817	BLIND	5619780	3	3	FLG COMPACT BLIND 1500# RJ ALLOY 625 B564 UNS N06625 PIPE CLASS 1500A60	0	0	
750										
750							*****ORIFICE FLANGES*****			
760										
760			ORIFICE FLG	5651201	2	3	WN ORIF SET 1500# RF ALLOY 825 B564 UNS N06625 SCH 40S	0	0	
770										
770							***** PIPE NIPPLES*****			
780										
780		814		0			NIPPLES BBETO BE CUT FROM PIPE			
790										
790							***** GASKETS*****			
800										
800	U04		GASKET	5673801	0.5	8	GASKET 1500# ALLOY 625 SPIRAL WOUND GRAPHITE IR	0	0	
810	U04	817	GASKET	5673460	3	8	GASKET 1500# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE LINE CLASS 1500A60	0	0	
820										
820		B1					***** BOLTING*****			
830										
830	U04	BA1	BOLTING	5675770	0.5	3	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
840	U04	BA4	BOLTING	5675776	0.5	3	STUD-BOLT WITH 2 NUTS API-6A718	0	0	
850										
850							***** LINE BLANKS*****			
860										
860			SPEC BLIND	5750310	0.75	3	SPECTACLE BLIND 1500# RF ALLOY 625 B443 UNS N06625	0	0	
870	U04	817	SPEC BLIND	5750411	3	3	SPECTACLE BLIND COMPACT 1500# RJ ALLOY 625 B443 UNS N06625	0	0	
880										
880	U04	817	PADDLE	5750290	3	3	PADDLE BLANK COMPACT 1500# RJ ALLOY 625 B443 UNS N06625	0	0	
890	U04	817	PADDLE	5750291	3	3	PADDLE SPACER COMPACT 1500# RJ ALLOY 625 B443 UNS N06625	0	0	
900										

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD	CORR. ALLOW. 1 MM	SOUR SERVICE NACE STD. MR0175 / ISO 15156 DESIGN: 184 BARG AT 110°C CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: INCONEL 625 FLANGE RATING: 1500 RF/CF TEMPERATURE RANGE: -100 / 110 °C	12/16/2024	BY OZAC	U04	Page 2 of 2	1500A60